

09 — Unrealizable Specifications

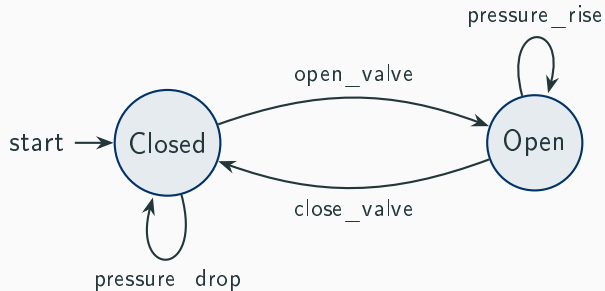
Diagnosis and Counterexamples

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Mununu Tutorial

The Valve Controller

A simple valve with two states and four events:



- **Controllable:** `open_valve`, `close_valve` (controller chooses).
- **Uncontrollable:** `pressure_rise`, `pressure_drop` (environment).

The Automaton in CTXDSL

```
1      automaton Valve {
2          controllable {
3              label open_valve;
4              label close_valve;
5          }
6
7          states {
8              state Closed initial;
9              state Open;
10         }
11
12         transitions {
13             transition Closed -> Open on label open_valve;
14             transition Open -> Closed on label close_valve;
15             // Environment events
16             transition Closed -> Closed on label pressure_drop;
17             transition Open -> Open on label pressure_rise;
18         }
19     }
```

An Impossible Specification

```
1  mu_formulas {
2      // UNREALIZABLE: demands being in both Open and Closed simultaneously.
3      // No state satisfies (Open && Closed) since they are distinct states.
4      formula impossible_spec {
5          over Valve;
6          body = Open && Closed;
7      }
8
9      // REALIZABLE: the valve can reach Open state from any state.
10     formula open_reachable {
11         over Valve;
12         body = mu X. (Open || <> X);
13     }
```

Open && Closed — no state is simultaneously Open *and* Closed.

Unsatisfiable (0/2 states).

$\mu X. (Open \vee \hat{\vee} X)$ — the valve can reach Open. Satisfiable (2/2 states)

Controller with Diagnostics

```
1  controllers {
2      // This controller will be UNREALIZABLE.
3      // Diagnostics reveal why: no initial state satisfies the spec.
4      controller impossible_valve {
5          source Valve;
6          satisfying impossible_spec;
7          diagnostics {
8              counterexample = true;
9              deadlock_traces = true;
10         }
11     }
```

Running synthesis with diagnostics enabled:

```
$ mununu context synth 09_unrealizable.ctxdsl \
  --controller impossible_valve
Controller impossible_valve: UNREALIZABLE
Violating initials: {Closed}
```

Takeaway

Concept	Meaning
Unrealizable	No controller can enforce the specification — the property is too strong or contradictory.
<code>diagnostics</code>	Enables detailed output explaining failure.
<code>counterexample</code>	Shows a concrete witness of why synthesis fails.
<code>deadlock_traces</code>	Shows traces leading to states where the controller gets stuck.

Best practices:

- Always include a **realizable** formula for contrast (e.g., safety invariant).
- Enable diagnostics early — they are your debugger for specifications.
- An unrealizable result is not a tool failure; it means the spec needs revision.